

Multi-Camera Pedestrian Detection, Tracking & Re-Identification using Joint ConvNeXt V2 Architecture

A compact system that carries a person from synchronized camera pixels to reviewable local tracks and ranked cross-camera candidate evidence.

62.8%

Cross-camera Rank-1

P-DESTRE validation - Protocol D

64.08

MOT17 MOTA

Val-half - Protocol A

7.78M

Model parameters

Approx. 18 FPS full pipeline

256-D

Identity descriptor

RoIAlign and four body strips

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Software: Apache-2.0

P-DESTRE-derived demonstration media: CC BY-NC-SA 4.0

A shared feature hierarchy connects perception to analyst review.

ConvNeXt V2 produces detail-rich P2 and context-rich P3 features. A CenterNet branch decodes pedestrian boxes while a part-aware branch extracts a normalized identity descriptor from each detected region.

01 Camera input Four synchronized views, letterboxed to 1088 x 608.	02 ConvNeXt V2 Shared P2/P3 features preserve edges and semantic context.	03 CenterNet Heatmap, box, offset, and IoU quality outputs.	04 Part descriptor RoIAlign, four strips, Coordinate Attention, 256-D fusion.	05 BoT-SORT Motion, overlap, and appearance maintain camera-local tracks.	06 Identity review Cross-camera gallery ranking exposes evidence for human review.
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Detection objective $L_{det} = L_{focal} + \lambda_{box} L_1 + \lambda_{iou} L_{GloU}$ Center heatmaps and decoded boxes share the same fused feature map.	Part fusion $z = \text{normalize}(\sum \alpha_k z_k)$ Coordinate Attention weights the evidence retained by each body strip.	Gallery similarity $s(q,g) = \text{weighted cosine}(z_q, z_g)$ Similarity ranks candidates; it is not a statement of identity.
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Detection, tracking, retrieval, and efficiency use separate protocol labels.

Evaluation	Retrieval	Detection	Tracking
P-DESTRE fold-0 validation - Protocol D	62.8% Rank-1	90.74% mAP@0.5	-
P-DESTRE fold-0 test - Protocol E	61.3% Rank-1	88.4% mAP@0.5	-
MOT17 val-half - Protocol A	-	-	64.08 MOTA / 74.24 IDF1 / 61.34 HOTA

System profile	Parameters	Input	Descriptor	Throughput condition
EffiPed Tier-1 / G_tier1	7.78M	1088×608	256-D	Approx. 18 FPS - full pipeline - NVIDIA RTX 5080 Laptop GPU

Interpretation

These values come from the checked-in aggregate evidence fixture. The hosted replay is an application demonstration and is not used to compute the benchmark cards. Performance is sensitive to occlusion, localization quality, pose, illumination, camera topology, time gaps, and domain shift.

The browser replay mirrors the original PedestrianTracker workflow.

Demo evidence	Value	Role in the interface
Synchronized views	4	Switchable four-camera playback
Processed frames	150	Frame stepping and seek timeline
Camera-local tracks	367	Detector and tracker annotations
Cross-camera IDs	45	Ranked gallery evidence
Indexed query examples	3	Query selection and candidate inspection

Hosted demo Precomputed and Vercel-safe The site serves two optimized archived replays, query crops, and gallery candidates. The video already contains the tracker-rendered boxes; the browser does not fabricate or reposition detections.	Local mode FastAPI and CUDA inference Uploads, job progress, search-by-example, asset retrieval, and explicit cleanup are available when an authorized checkpoint is mounted locally.	Related research PartJDE and BoxJDE Matched PartJDE readout gain: +6.66 pp Rank-1. BoxJDE source-level gain: +13.64/+12.94 pp Rank-1/mAP.
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Responsible-use boundary

The system is designed for user-authorized, human-in-the-loop review. Candidate similarity is not identity proof. Public checkpoints remain withheld until all training-data redistribution terms are confirmed.

References: P-DESTRE - arxiv.org/abs/2004.02782 | Media license - creativecommons.org/licenses/by-nc-sa/4.0/ | Source - github.com/aswanth-07/effiped-multi-camera-tracking